Team 7

JAMN INC.

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Part 2: Prototype Documentation

I. System Description

a. User Narrative + Key User Tasks

Molly, a 60 year old woman, heard how her friend got scammed recently and wants to know what to look out for. She looks up on her laptop web browser a game called "Phish or Fail." When she clicks on the search she is brought to the home screen where she is presented with 3 simple options: Play, Learn, and Settings (Screenshot 1). Since she doesn't know what to look for she goes to the learn page where she can learn about the red flags, what to do in response, and the types of scams under 3 categories (Phone, messages, and email)(Screenshot 2). She reads through the learn page and even clicks through the quick tip cards and feels prepared to play the game and clicks "Play Game" at the bottom (Screenshot 3).

Now she sees the set up page for scenarios, but she wants to make sure she's on the easy mode to start out with as she is only a beginner (Screenshot 4). She clicks the settings button in the top right corner and makes sure that hints are enabled and the game is on easy mode when she saves the settings that's confirmed with a message and clicks the back button (Screenshot 5). She wants to see if she can win as a target and thinks to herself how cute the little fish is. So she chooses the target option for the role and knows that she has to deal with emails all day so the ensnaring emails option would be familiar to start with (Screenshot 4).

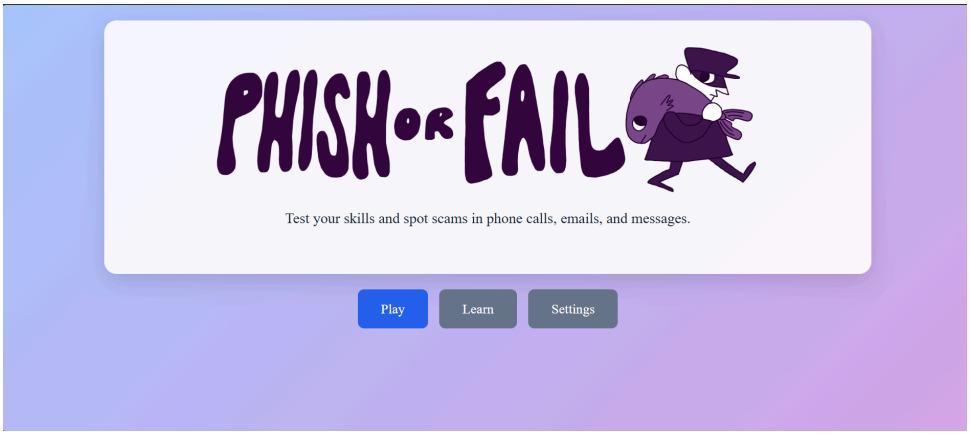
Upon choosing "Start Game" she reads through the email and determines that it was a phishing attempt and not a legitimate email (Screenshot 6). She chooses correctly and gets shown a page that outlines why it was a phishing email and she reviews the little signs that were off. She wants to try again but this time as an attacker (Screenshot 7). So she clicks "Try Again" but changes to the attacker in the set up page (Screenshot 4). She is given a prompt to choose options to build a phishing email (Screenshot 8). She has to decide which are the best choices for the email, subject, opening, content, and link (Screenshot 8). She thinks this is really neat because she has never built an email like this before. Once she submitted her build she failed because she got an 80/100 because her build was too weak due to her opening line (Screenshot 9).

Molly finds this game fun with the scenarios it offers and thinks of who she can recommend the game to. One person that comes to mind is her 11 year old grandson and thinks this would be a good learning tool because he just got an iPhone this year. Teaching him about what to look out for would be imperative so he doesn't fall victim due to his lack of knowledge. She also thinks that her boss would find it a fun game to do as team refresher or competition at work due to the cyber security issues the company has been having. She always says you can never be too prepared and at least it's not another 3 hour long training so that will keep her coworkers engaged. She also thinks of the places she volunteers at for the underprivileged community and believes that this could be a great soft introduction to learning more digital literacy since it's in a game format and there are visuals. In addition, since there are more visuals she feels that her

friend who has English as a 3rd language could still use it due to the intuitive design with the colors of buttons and the fun pictures associated with many of the actions. Finally, she wants to introduce the game to her granddaughter who has become a “micro influencer” who’s social media presence makes her a target for scams, and she can appreciate the game’s aesthetic since she’s color blind.

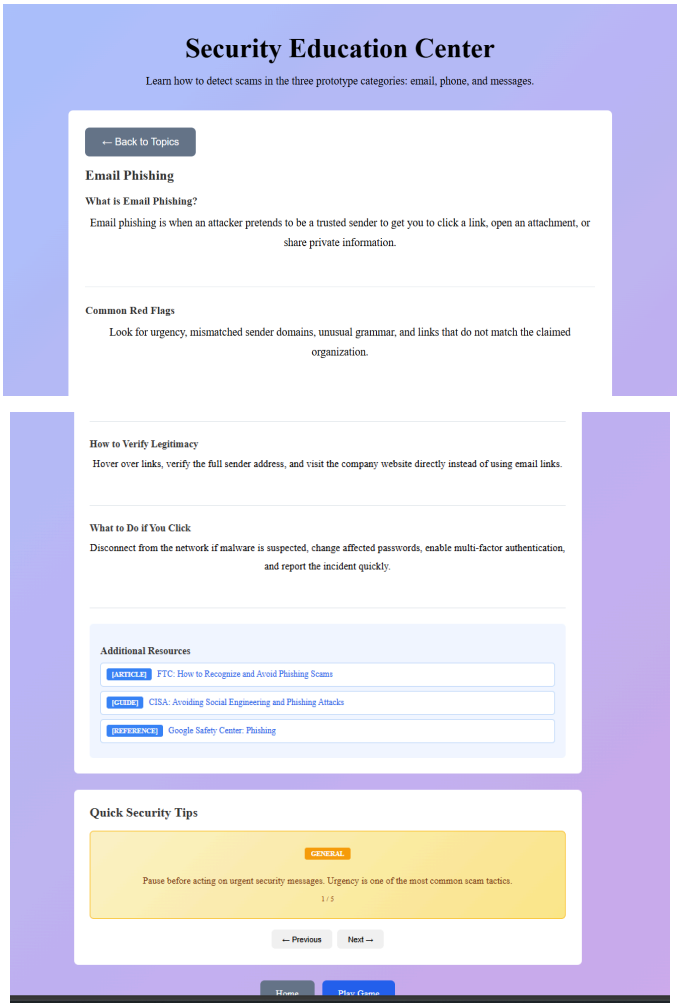
b. Screenshots + Descriptions

Home Screen when clicking on the Website (Screenshot 1)

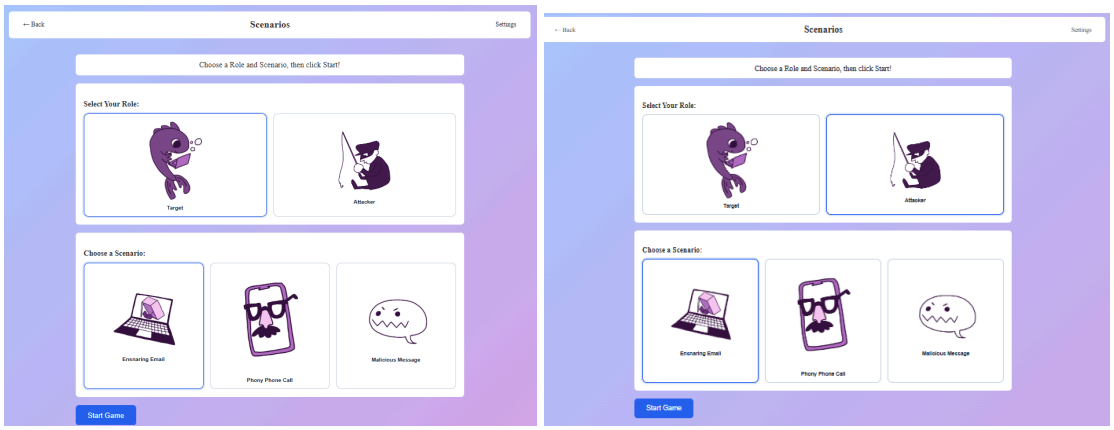


Learn Page General with Security Tips (Screenshot 2)

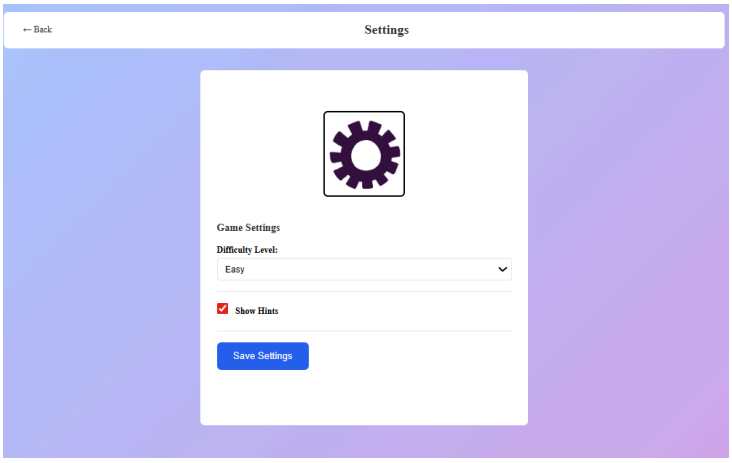
Learn Page with more information on specific category: Email (Screenshot 3)



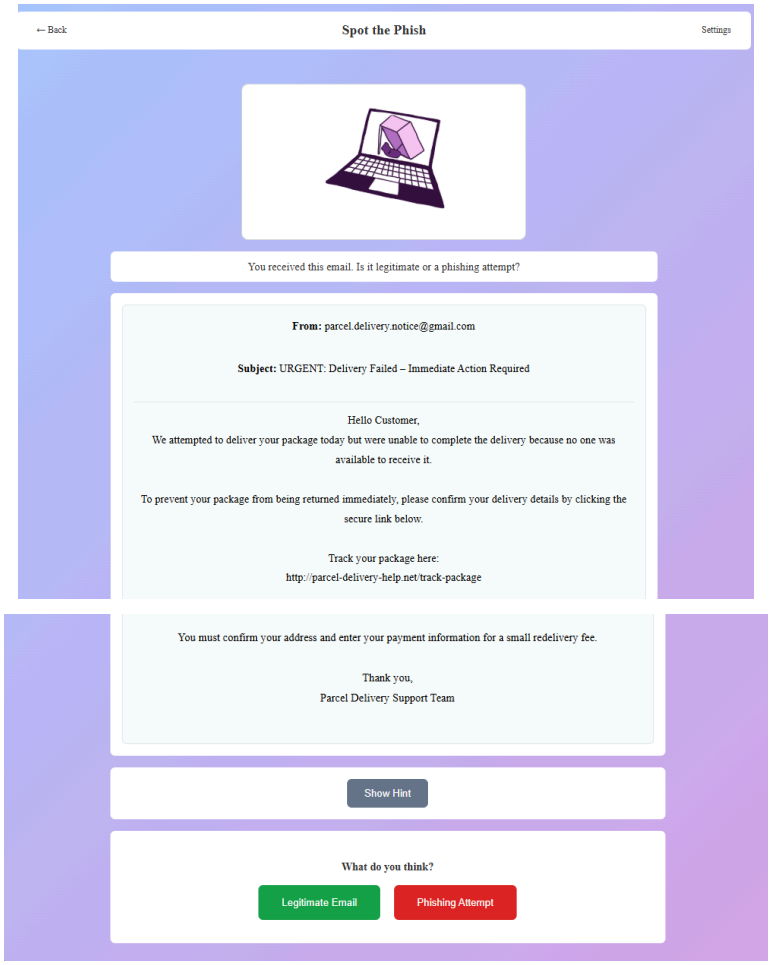
Scenario Page after “Play Game” or “Try Again” is selected, can Choose Role & Scenario (Screenshot 4)



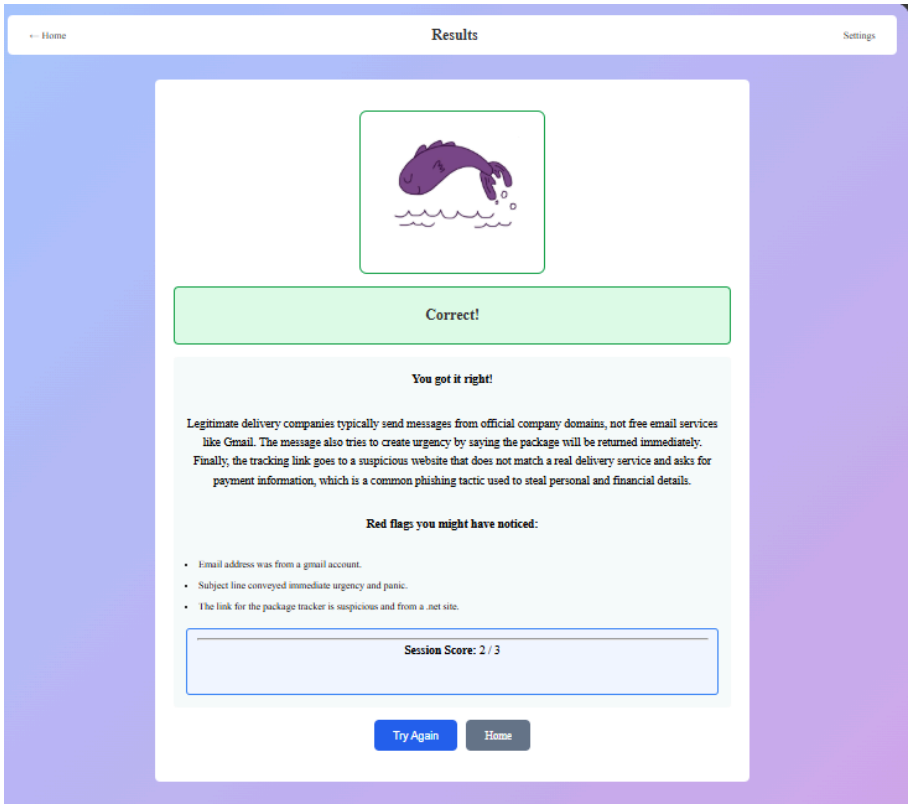
Settings Page (Screenshot 5)



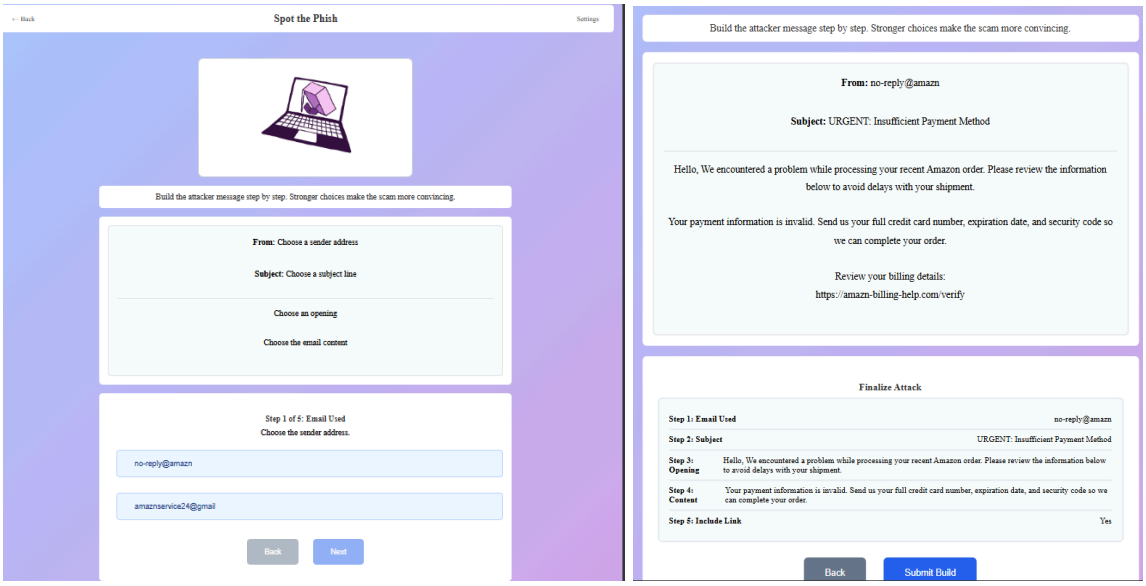
Phishing Email Easy Mode Question (Screenshot 6)



Correct Answer for Phishing Email Easy Mode Question (Screenshot 7)



Individual Prompt for Sender Address and Finalized Build for Easy Mode (Screenshot 8)



Result Page for Build that Failed Easy Mode Prompt (Screenshot 9)

II. Design Justifications

a. Why "Phish or Fail!"?

Between the three designs from Milestone 2, "Phish or Fail", "SecureKey", and "Wallet Wingman", "Phish or Fail" was the chosen design. After reviewing the trade-offs of the different designs, "SecureKey" was deemed unfeasible to implement, and "Wallet Wingman" exhibited redundant characteristics. The evaluation metrics would also have been difficult to decide on. "Phish or Fail!" provided a low-stakes, yet educational and engaging experience for users. It also had the greatest potential to explore factors that could improve the human-computer interaction experience and to measure and evaluate success. The simulated backend would allow different scenarios to be shown to the user, enabling them to engage with a functional application. This design choice also allowed for a unique spin on financial security games. From the few that do exist, not many allow the user to play as the attacker, giving a new perspective on the topic. Making a game provides an easy way to learn about a more difficult or stressful topic. The proactive approach of this system places users in a fraudulent environment, but in a way that does not actually put them at risk or cause any harm.

b. Design and Relation to Problem

A web application was chosen because it would be the most accessible for all stakeholders, older folks who have trouble with an iPhone wouldn't be able to use tools only on a phone well.

Meanwhile, a child may not have access to a phone yet from their parents, but can use a library computer or a home computer which is why a mobile app would not be the best option. In addition, the web application can be widely distributed for a large group of people to go to without having to do the hassle of downloading an app and signing in, it would be a very quick start up time of use.

In relation to the problem of lack of knowledge surrounding phishing scams, this game is an engaging way for users to improve their knowledge without sitting through a monotonous training video or reading article after article. This design allows the user to test their skills in a simulated environment to determine if they are prepared enough to recognize actual phishing attempts. If they feel they need more knowledge there is a learn page where they can see tips, and common red flags as well as articles that can supplement their reading. It's better for their overall skills to have that knowledge in the back of their head instead of another 2 factor authentication or another budgeting app. A main source of security breaches is employees getting phished even with all the cyber security in place. Reducing human error and raising the level of awareness is one of the best options for financial security.

c. Making an Informed Decision

Data from Milestones 1 and 2.

In Milestone 1, our team conducted interviews and reviewed studies that showed users are concerned with financial awareness and financial safety. Users are concerned that they lack a high-level understanding of how to best keep their financial information secure. One interviewee in particular shared a lot of concern regarding her elderly mother's fear of using the internet because of scammers, and wished there was a way to educate her more on the matter, so that she would be less anxious about going online. We also looked at some preexisting solutions for this problem as we researched and found that the games did not seem to be substantially educational in our opinion, which gave us a desire to make something better. Seeing those examples helped us to find a balance between game and education when later implementing the program.

In Milestone 2, we researched real examples of phishing emails when creating the mockup and storyboards for Phish or Fail. We wanted our answers to model what users might actually see in a real scam email scenario, so using the real data we found, we created mock options for the game. We continued this design decision in Milestone 3 as well when we began adding different scenarios. Having these real examples also helped us to design our scoring system and the remaining scenario-based game structures.

Studio 2 Insight and Feedback

In Studio 2, our peers offered some feedback that we took into consideration as we chose and implemented one of the designs we presented. The main reasons we did not proceed with SecureKey was because it was deemed during Studio 2 to be unfeasible given the nature of this

project and also received generally negative feedback as well, as some thought it was redundant within the problem space and an unnecessary accessory to carry around. Regarding Wallet Wingman, our peers thought that it was interesting, but also fell into an oversaturated space. Additionally, it was thought to be very difficult to implement. Phish or Fail seemed to make peers more excited, as they offered examples and showed the most interest in its design and the future of its implementation. Peer excitement made us confident that this design in particular was the one we should proceed with. We looked at the examples given to us by peers, but opted not to model our game like them, as we wanted it to be as fun and original as possible. Overall, it was extremely helpful to hear what our peers thought and made the decision to move forward with Phish or Fail easy with their support.

User Research Findings.

Our user research highlighted a widespread lack of financial awareness across diverse populations in the United States. Despite the significant role finances play in shaping life outcomes, less than half of college students have received formal education in personal finance, which is suggestive of an even larger gap among those outside higher education. Through conversations with peers and observations in class, we found that many college students feel unprepared to manage their finances and would benefit from accessible educational tools. Research also revealed that this issue extends beyond students, affecting middle-aged and older adults who seek better financial knowledge for long-term planning, such as retirement. Certain groups such as young adults, women, individuals without a high school diploma, unemployed individuals, and racial minorities are particularly vulnerable to financial illiteracy. Overall, our findings confirmed that financial awareness is a critical and widespread problem, reinforcing the need for inclusive and accessible solutions.

When designing a program, it is incredibly important to design with consideration for a variety of users with diverse backgrounds, accessibility needs, and digital literacy, as certain populations can be more vulnerable based on these attributes. For older adult users who might not be as "tech-savvy" we made sure that the navigation was simple and straightforward and had consistent interaction patterns to reduce confusion and cognitive load. This allows them to be confident while navigating the system as they learn. Next, we wanted to ensure some accessibility needs from the beginning, such as choosing a blue-violet color palette and including images that can communicate without colors, as well as selecting an easy to read font and black text. We also took into consideration neurodivergent users by making sure the interface was not overstimulating and the application had a predictable layout and cool-tone colors, as well as clear user feedback and resources to learn more to prevent users being overwhelmed and/or confused. Additionally, we took into account users who might not speak English as their first language or users from under-resourced communities who might need clear instructions in simplified language, as well as visual cues to assist their navigation and decision making. We made sure

instructions felt uncomplicated and relied on number-based grading for communication across language and education barriers. Another vulnerable user group we found in our research were social media users, as many scams happen via social media. To draw users from that demographic we took into account aesthetics, and made sure that the simple platform and visual aids were both effective and attractive so that users could possibly be drawn in by looks as well as functionality and education. Ultimately our findings helped us to approach the design with a lot of care and guided us to implement a program that can effectively be utilized by many different people.

III. Implementation Challenges

a. Technical and Design Challenges

Project development posed numerous technical and design challenges. Due to the scope of the assignment and less experience with frontend development, there was a learning curve. Drawing on previously created projects and basic game development knowledge, GitHub Copilot played a helpful role in developing a functional prototype. There were many factors to consider to ensure consistent behavior, such as game progression, scoring, and hints. It proved challenging to balance the educational aspect with having fun while also considering the text, its length, and more, so as not to overwhelm or deter the user. It was also more difficult than originally expected to create the scenarios, having to make them unique yet understandable, requiring thinking, but not overly difficult. Creating a responsive design was difficult, as ensuring all elements worked and were responsive across different screen sizes was challenging. The logic of the attacker role was also confusing to implement and maintain consistency and accuracy, and GitHub Copilot was a helpful resource.

b. Intended Features, Not Implemented

Throughout the development process, we focused on core functionality and creating a usable application. However, we intended to build some features but could not, some due to time constraints and others due to complexity and scope. One feature was user accounts, which allowed for saved progress and level progression throughout the game. Similarly, as the user progresses, the difficulty would update automatically based on their performance and score. Instead, we decided to allow the user to choose between three levels and then mitigate difficulty through the hint system. Additionally, we would have added more gameplay categories, along with a wide variety of scenarios, but it would have been quite extensive and time-consuming. All scenarios were created uniquely by the team, and it would have been a toll to create numerous different questions. Additionally, due to limited experience with user interface and game design, we would have implemented a drag-and-drop feature for the attacker roles, along with the ability to create the player's own scenario. This would have been particularly helpful if the user were an instructor, using the application as an educational tool. Other features that were not implemented included options typically available in other games or websites, such as language selection, text

scaling, or text-to-speech. Other features included a timed mode for the game that simulated real life, a voice-call simulation, and a multiplayer mode.

c. Cause of Limitations

Firstly, time constraints limited the scope to learning outcomes and core gameplay first. Given the limited development timeline, we chose to prioritize those two aspects. Additionally, creating the educational, research-based scenarios and then designing games surrounding them took more time than expected, so there was less time to work on advanced features. The prioritization of core functionality and stability was also influenced by the limited team bandwidth, as we had to work with our limited resources, experience, and personnel. This resulted in the project being front-end focused, so backend features such as profiles, authentication, and cloud analytics were put on hold. Testing resources were also constrained, so we were less able to validate edge cases across various browsers and devices.

d. Future Plans and Implementation

Given more time, we would first focus on implementing a backend for authentication and profile storage so that users can sign in and view their tracked progress. With the addition of profiles, we would build a richer results dashboard with historical performance and targeted tips for improvement. Next, we would expand the scenarios and categorize them by threat type, confidence level, and difficulty. We would also improve accessibility in order to reach more users and meet stronger compliance requirements, such as ARIA labels, keyboard navigation, contrast tuning, and screen readers. We would also conduct wider usability testing, measuring user behavior and recording feedback in order to fine tune the application. Within the scenarios, we would love to implement adaptive learning logic to personalize hints and the difficulty of tasks.

IV. Technology Justification

To develop the system prototype, HTML, CSS, and JavaScript were used to build a browser-based platform that would support the application's usability goals and evaluation needs. A web application meant users could interact with any device with a browser, without having to install new software or dependencies. It enhances ease of use and accessibility, enabling straightforward system interaction. The chosen framework and language provided the freedom to design the application. It enabled the development of a simple, functional, user-friendly game that would provide a smooth flow and a positive learning experience. CSS made the application robust and adaptable, enabling the creation of a visually appealing interface. The use of a mock backend and heavily focused front-end allowed development to focus on game flow, factors that influence human-computer interaction, and functionality. It was easy to include performance metrics, such as the score tracker, and observe how the user would interact with the game. It was straightforward for usability evaluation. C++ and Java were avoided because it would have required a deeper focus on memory management. Python would have required dependencies and

would not have been the best option for an appealing interface. Game engines like Unity would have been time-consuming and overall heavy, resulting in inefficient development. HTML, CSS, and JavaScript provided an easy setup, easy to share, a fast build, and an effective demo.

Part 3: Initial Usability Specifications and Evaluation Plan

Evaluation Plan Overview

To evaluate the usability and effectiveness of the Phish or Fail prototype, we will conduct a usability study with participants who are representative of our target user group. Participants will be asked to complete a series of core tasks, including starting the game, selecting scenarios, completing both target and scammer activities, and reviewing feedback.

During testing, both quantitative and qualitative data will be collected. Quantitative data will measure performance through metrics such as time on task, error rate, and task completion rate. Qualitative data will be gathered using think-aloud protocols, observation, post-interviews, and surveys to better understand user behavior, engagement, and perceptions.

This mixed-methods approach allows for a comprehensive evaluation of both system performance and user experience.

I. Revisiting Milestone 1

In Milestone 1, the tasks defined by our team were much broader in relation to the domain of financial awareness. Some tasks stated include:

1. Make informed decisions that protect financial safety
2. Gain awareness of common fraud schemes and techniques
3. Recognize and report fraudulent activity
4. Minimize damage from scammers or cyber attacks

These tasks correlate to our Phish or Fail prototype on a high-level, but do not cover the specific functions and goals of our game. Accordingly, we have created an updated task list rooted in these principles but more directly measurable as it relates to our system.

Task 1: Begin Game
Description: The user navigates the page to begin playing Phish or Fail.
Success: The user locates and initiates gameplay without confusion.
Quantitative Metrics for Measurement:

- Time on task
- Error rate

Qualitative Metrics for Measurement:

- Usability

Task 2: Select Role and Scenario

Description: The user chooses which role they want to play (scammer or target) and which scenario (email, message, or phone call).
Success: The user successfully selects both a role and scenario without hesitation or error.
Quantitative Metrics for Measurement:

- Time on task
- Error rate

Qualitative Metrics for Measurement

- Usability
- Confusion

Task 3: Complete Target Scenario
Description: The user navigates through a target scenario (email, message, or phone call) designed to see if they can recognize scams.
Success: The user finishes the scenario and recognizes the scam.
Quantitative Metrics for Measurement:

- Time on task
- Error rate (did the user recognize or not)

Qualitative Metrics for Measurement

- Usability
- Confusion

Task 4: Receive Score and View Feedback (Target)
Description: The user views the score on their target scenario and engages with feedback.
Success: The user understands the feedback provided by the game.
Quantitative Metrics for Measurement:

- Time on task

Qualitative Metrics for Measurement

- Usability
- Confusion
- Engagement
- Real-World Applicability

Task 5: Complete Scammer Scenario
Description: The user navigates through a scammer scenario (email, message, or phone call) designed to see if they can understand the components of a scam.
Success: The user successfully designs a scam.
Quantitative Metrics for Measurement:

- Time on task
- Error rate (did the user understand components)

Qualitative Metrics for Measurement

- Usability
- Confusion

Task 6: Receive Score and View Feedback (Scammer)
Description: The user views the score on their scammer scenario and engages with feedback.
Success: The user understands the feedback provided by the game.
Quantitative Metrics for Measurement:
- Time on task

Qualitative Metrics for Measurement
- Usability
- Confusion
- Engagement
- Real-World Applicability

Task 7: Navigate to Next Action
Description: The user is presented with the choice to try again or return to the home page.
Success: The user tries again or navigates to the home page.
Quantitative Metrics for Measurement:
- Time on task
- Error Rate

Qualitative Metrics for Measurement
- Usability
- Confusion
- Engagement (did the user choose to play again)

Task 8: Navigate to Learning Page
Description: The user navigates to the learning center.
Success: The user can view the learning page.
Quantitative Metrics for Measurement:
- Time on task
- Error Rate

Qualitative Metrics for Measurement
- Usability
- Confusion
- Engagement

II. Quantitative Benchmarks

Quantitative benchmarks are assigned to each task to create meaningful standards of performance [1]. "The following benchmarks define expected user performance for each task, allowing for objective evaluation of usability, efficiency, and learning outcomes."

Task 1: Begin Game

Specific Steps:

1. Open the application homepage
2. Locate the “Play” button
3. Click “Play” to begin the game

Benchmarks:

- Time on Task: ≤ 10 seconds
- Error Rate: ≤ 5%
- Completion Rate: ≥ 95%

Task 2: Select Role and Scenario

Specific Steps:

1. View available roles (Scammer or Target)
2. Select a role
3. View available scenarios (email, message, phone)
4. Select a scenario
5. Begin the scenario

Benchmarks:

- Time on Task: ≤ 15 seconds
- Error Rate: ≤ 10%
- Completion Rate: ≥ 95%

Task 3: Complete Target Scenario

Specific Steps:

1. Read the scenario content (email/message/phone)
2. Identify key elements of the interaction
3. Decide if the scenario is legitimate or a scam
4. Submit response
5. Proceed to feedback

Benchmarks:

- Time on Task:
    - Easy: ≤ 30 seconds
    - Medium: 30-60 seconds
    - Hard: 60-90 seconds
- Accuracy:
    - Easy: ≥ 85%
    - Medium: ≥ 75%
    - Hard: ≥ 65%
- Overall: ≥ 75% average

- Error Rate:
    - Easy: ≤ 15%
    - Medium: ≤ 25%
    - Hard: ≤ 35%

Task 4: Receive Score and View Feedback (Target)
Specific Steps:
1. View score/result
2. Read explanation of correct answer
3. Identify key indicators of scam or legitimacy
4. Continue to next screen

Benchmarks:
- Time to Interpret Feedback: ≤ 20 seconds
- Completion Rate: ≥ 95%
- Understanding Accuracy: ≥ 80%

Task 5: Complete Scammer Scenario
Specific Steps:
1. Review scenario instructions
2. Select components (sender, subject, link, etc.)
3. Construct scam scenario
4. Submit response
5. Proceed to feedback

Benchmarks:
- Time on Task:
    - Easy: ≤ 45 seconds
    - Medium: 45-75 seconds
    - Hard: 75-100 seconds
- Accuracy (Correct Components Selected):
    - Easy: ≥ 85%
    - Medium: ≥ 75%
    - Hard: ≥ 65%
    - Overall: ≥ 75% average
- Error Rate:
    - Easy: ≤ 15%
    - Medium: ≤ 25%
    - Hard: ≤ 35%

Task 6: Receive Score and View Feedback (Scammer)
Specific Steps:

1. View score/result
2. Read explanation
3. Identify strengths and weaknesses
4. Continue to next screen

Benchmarks:

- Time to Interpret Feedback: ≤ 20 seconds
- Completion Rate: ≥ 95%
- Understanding Accuracy: ≥ 80%

Task 7: Navigate to Next Action

Specific Steps:

1. View options (retry or return home)
2. Select desired option
3. Navigate accordingly

Benchmarks:

- Time on Task: ≤ 10 seconds
- Error Rate: ≤ 5%
- Engagement (Replay Rate): ≥ 60%

Task 8: Navigate to Learning Page

Specific Steps:

1. Locate "Learn" section
2. Click to access the learning page
3. Browse available content

Benchmarks:

- Time on Task: ≤ 15 seconds
- Error Rate: ≤ 5%
- Completion Rate: ≥ 90%

III. Qualitative Data Collection Standards

Qualitative data refers to information or concepts that are not directly measurable on a numeric scale [3]. Instead, user experience, perception, and behaviors are measured. Collecting and analyzing qualitative data is essential for understanding the user outside of rigid quantitative performance metrics.

*A. Types of Qualitative Data Collecte*d

1. User Understanding
    a. Ability to understand game instructions and objectives
    b. Clarity of feedback provided after each scenario
2. Usability

    a. User confusion or hesitation during tasks
    b. Ease of navigation between screens and features
    c. Overall intuitiveness of the interface
3. User Engagement
    a. Level of interest and interaction throughout gameplay
    b. Willingness to continue playing or try additional scenarios
4. Real-World Applicability
    a. User perception of how well the game reflects real-world phishing scenarios
    b. Perceived ability to transfer learned concepts to real-life situations
5. Design & Content Feedback
    a. Feedback on features
    b. Clarity on explanations
    c. User-perceived ease of use
    d. Overall satisfaction with interface design

These qualitative measures capture user experience while also informing designers and developers in refining the system.

B. How Qualitative Data Will Be Collected

1. Think-Aloud Protocol: Participants test the system and verbalize all their thoughts as they move through the system [2].
2. Observation Notes: Participants move through the tasks and the facilitator takes notes on behaviors, including hesitation, confusion, and navigation patterns.
3. Post-Interview: After participants use the system, they answer brief, semi-structured follow-up questions about the user experience, focused primarily on design decisions.
4. Survey Responses: Participants will complete a post-test survey with open-ended questions regarding usability, satisfaction, and overall experience.

C. Justification of Method for Qualitative Data Collection

A. Think-Aloud Protocol

This method provides real-time insight into users' thought processes, allowing designers to identify confusion, misunderstandings, and cognitive load during interaction.

B. Post-Interview

Interviews allow for deeper exploration of user preferences, frustrations, and overall impressions, providing context that may not emerge during task completion.

C. Observation Notes

Observational data captures subtle behaviors (e.g., hesitation or repeated actions) that users may not explicitly report, helping identify usability issues.

D. Survey Responses

Surveys provide structured feedback on user satisfaction and usability. Additionally, written responses may encourage more honest or detailed feedback compared to verbal responses.

References

[1] Kate Moran. 2020. Benchmarking UX: Tracking Metrics. *Nielsen Norman Group*. Retrieved from https://www.nngroup.com/articles/benchmarking-ux/

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[3] 2021. Qualitative Data. *NNLM*. Retrieved from https://www.nnlm.gov/resources/data-glossary/qualitative-data